

Midnight Buildathon · Wave 1

One pledge per invoice. No lender sees another's book.

A registry on Midnight that refuses to let the same receivable be financed twice, while each lender sees only its own deals.

The problem

The same invoice, pledged to two lenders. Both pay out.

Lenders can't compare notes

Catching a double pledge means pooling books. A lender's clients and pricing are its business, so nobody shares.

Hash registry: bypassed

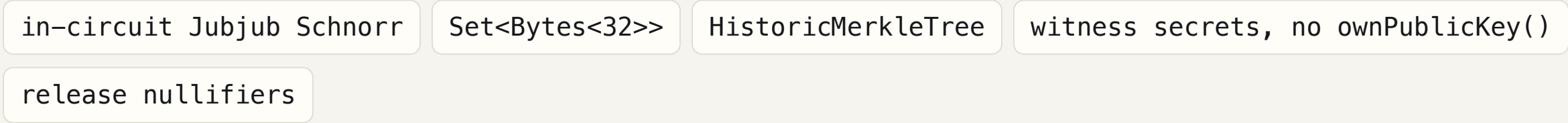
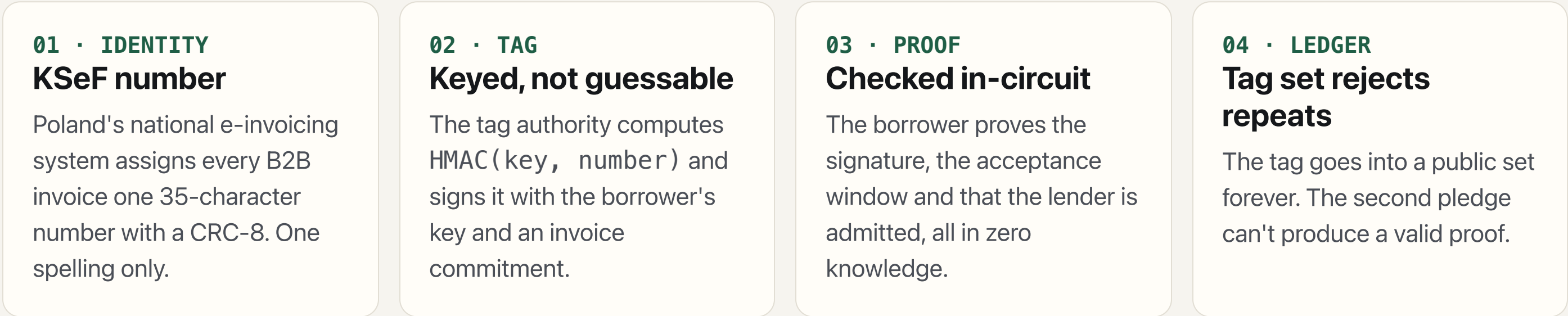
FV/2026/09/0412 and FV-2026-09-0412 hash differently. Each bank's spelling is a "new" invoice.

Hash registry: leaks

Anyone holding an invoice hashes it and learns from the public chain whether, and when, it was financed.

How it works

A canonical ID, a keyed tag, and a proof checked on Midnight.



Selective disclosure

Everyone learns exactly what they need.

PARTY	LEARNS	NEVER LEARNS
Financing lender	full invoice checked against the signed commitment	Other lenders' deals
Rival lender	"already pledged"	By whom, how much, which terms
Public chain	Opaque tag, note commitment, counts	invoice, amount, borrower, lender
Tag authority	Which invoices it attested	Which lender financed them

Pinned by privacy-leak tests that search the full public state and every circuit transcript for each private value.

A working registry, tested like it handles money.

4

circuits compiled with proving keys

95

tests: adversarial witnesses, forgery,
privacy leaks

3

surfaces: web demo on real circuits, CLI,
live indexer reader

Proven

- Five double-pledge variants rejected
- Ten forgery attempts rejected, including a malicious prover
- Stolen attestations can't be used by anyone else

Honest limits

- Single tag authority (threshold in Wave 3)
- Synthetic KSeF numbers until the test-environment adapter
- Facilities, draws and Lace UI are Wave 2

Start where every invoice already has one ID.

Beachhead: factoring on KSeF

Mandatory national e-invoicing gives every Polish B2B invoice a canonical number. Buyers: factors, supply-chain-finance banks, private-credit funds. Distribution: e-invoicing access points that already connect everyone.

Useful with just two lenders. No consortium needed.

Next

- **Wave 2:** Lace-connected borrower desk and lender inbox on Preprod, facilities and borrowing-base draws, encrypted invoice delivery, KSeF test-environment adapter, practitioner interviews
- **Wave 3:** threshold tags (no single party can tag), pledge assignment, reusable uniqueness module for any guessable-ID registry